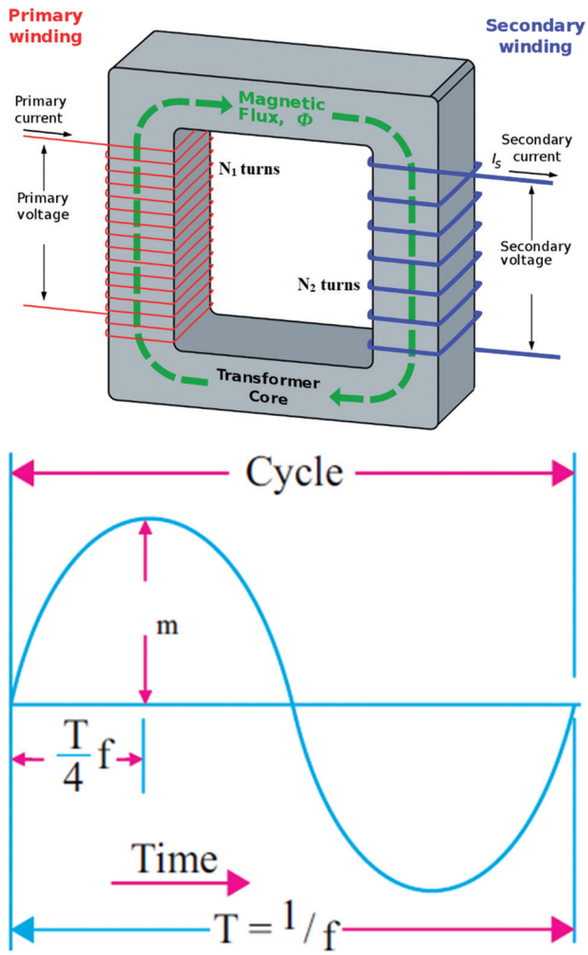


Fig 6.7 *Single Phase Transformer and Wave Form*

No. of turns in primary winding = N_1

No. of turns in secondary winding =N_2

Maximum value of flux = $Ø_m$ webers

Frequency of A.C input = f in Hz

The flux in the core will vary sinusoidally as shown in the above fig 6.7.

The flux in the core increases from zero to a maximum value in one quarter cycle (1/4f second)

i.e, Average rate of change of flux $= \frac{Øm}{\frac{1}{4f}} = 4fØm$

i.e average E.M.F induced per turn = 4f $Ø_m$ volts.

The flux varies sinusoidally. Hence the, R.M.S value of induced voltage is obtained by multiplying the average value by form factor which is equal to 1.11 for a sine wave.

i.e, R.M.S value of induced E.M.F per turn= 1.11 × 4f Φm volts.

= 4.44 f Φm volts.

The primary/secondary windings have N_1/N_2 turns respectively,

R.M.S value of induced e.m.f in primary E_1=4.44 f Φm N_1 Volts

R.M.S value of induced e.m.f in secondary E_2= 4.44 f Φm N_2 Volts

6.4.1 Voltage Transformation Ratio (k)

The ratio of secondary voltage to primary voltage is called voltage transformation ratio. It is represented by 'k'.

$$\frac{E_2}{E_1} = \frac{V_2}{V_1} = \frac{N_2}{N_1} = k$$

6.4.2 Current Ratio

By neglecting the losses

Input volt ampere = output volt ampere

$$V_1 I_1 = V_2 I_2 \quad \text{or} \quad \frac{V_1}{V_2} = \frac{I_2}{I_1}$$

$$\frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{1}{K}$$

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